

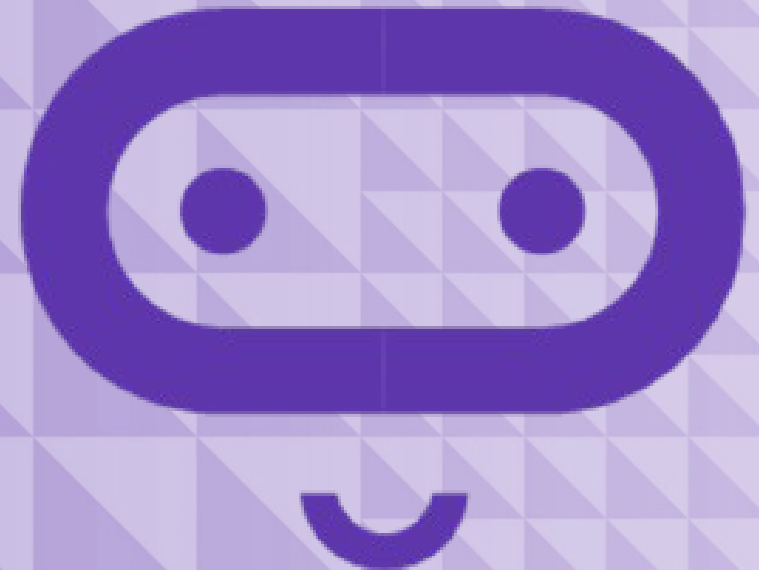
micro:bit

Introduction to Computer Science

Unit 1: Making with micro:bit

Lesson A

The micro:bit is for making

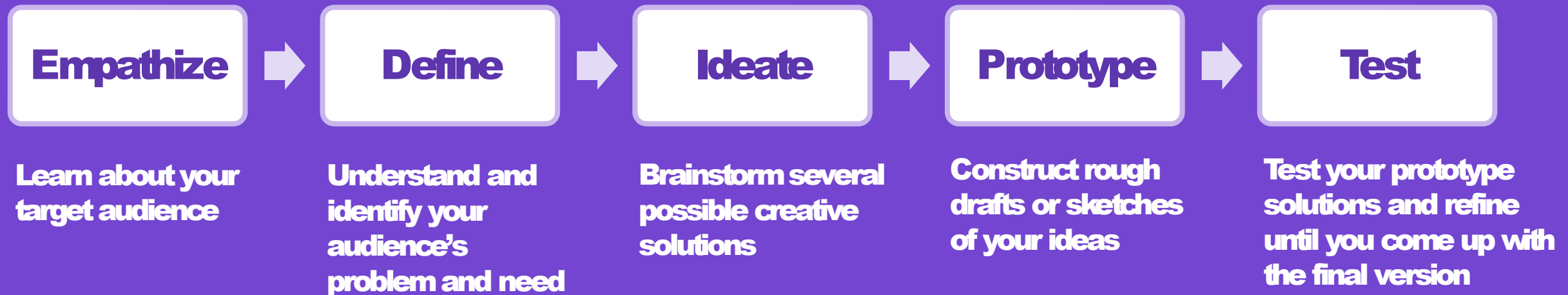


What we'll learn in Lesson A

- Design thinking/coding by design
- Interview a partner to imagine an ideal pet
- Create a definition statement and draw prototypes

Design thinking process

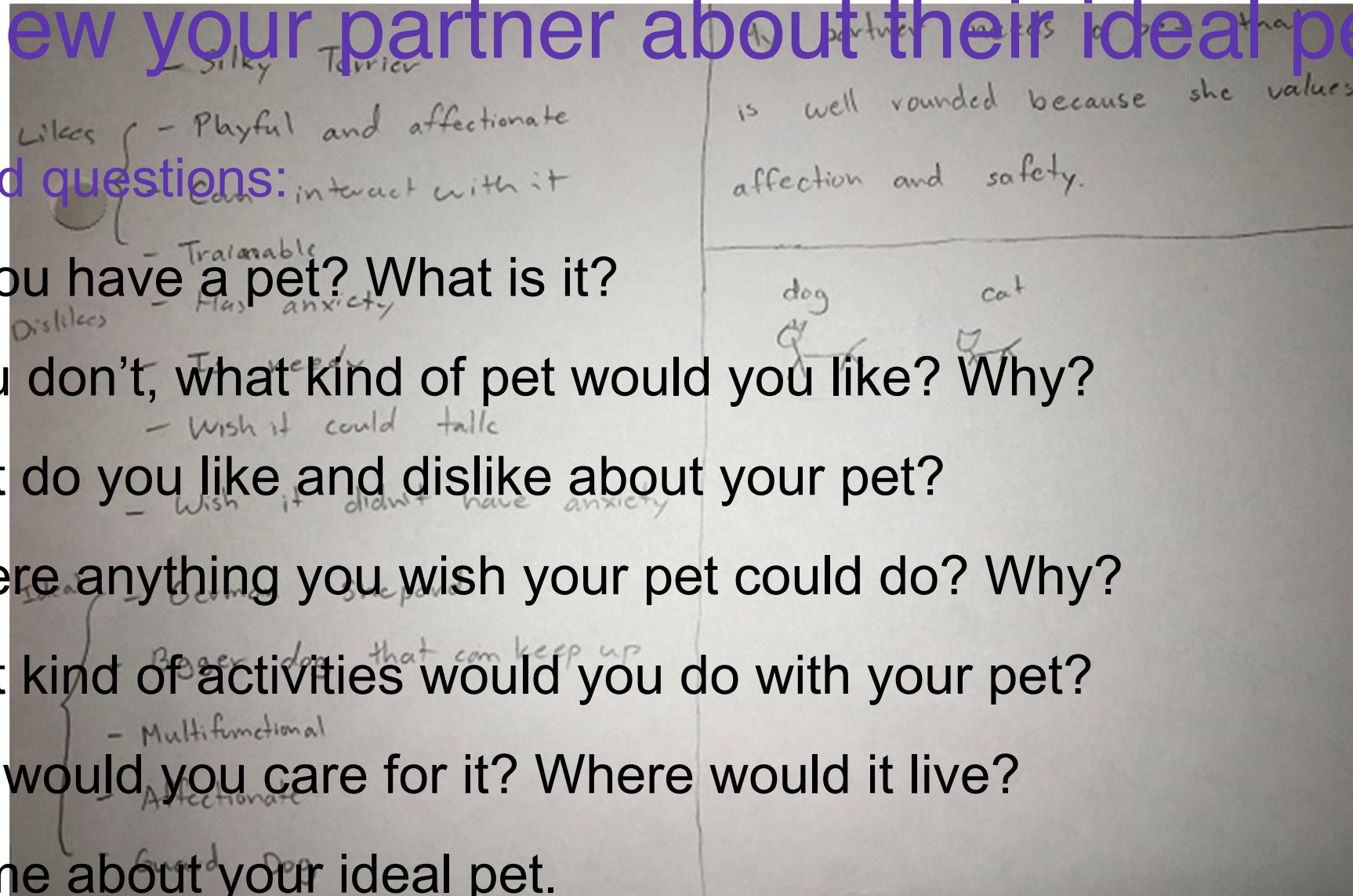
Whatever you build with code should serve a purpose or fill a need.



Interview your partner about their ideal pet

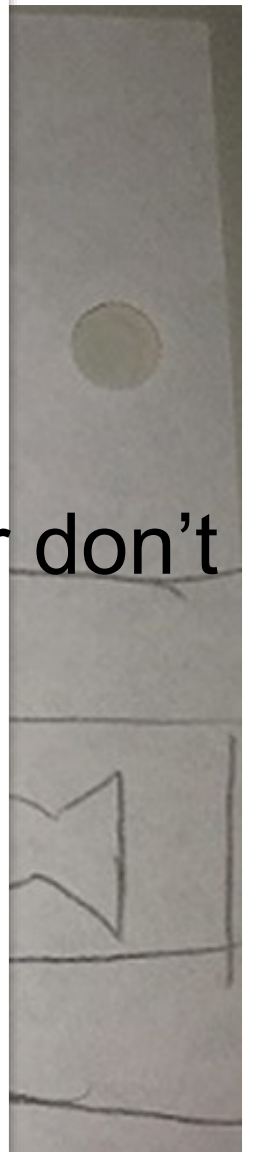
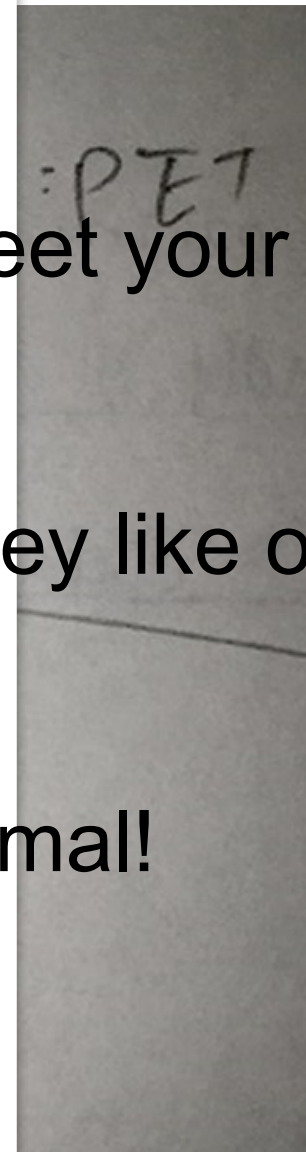
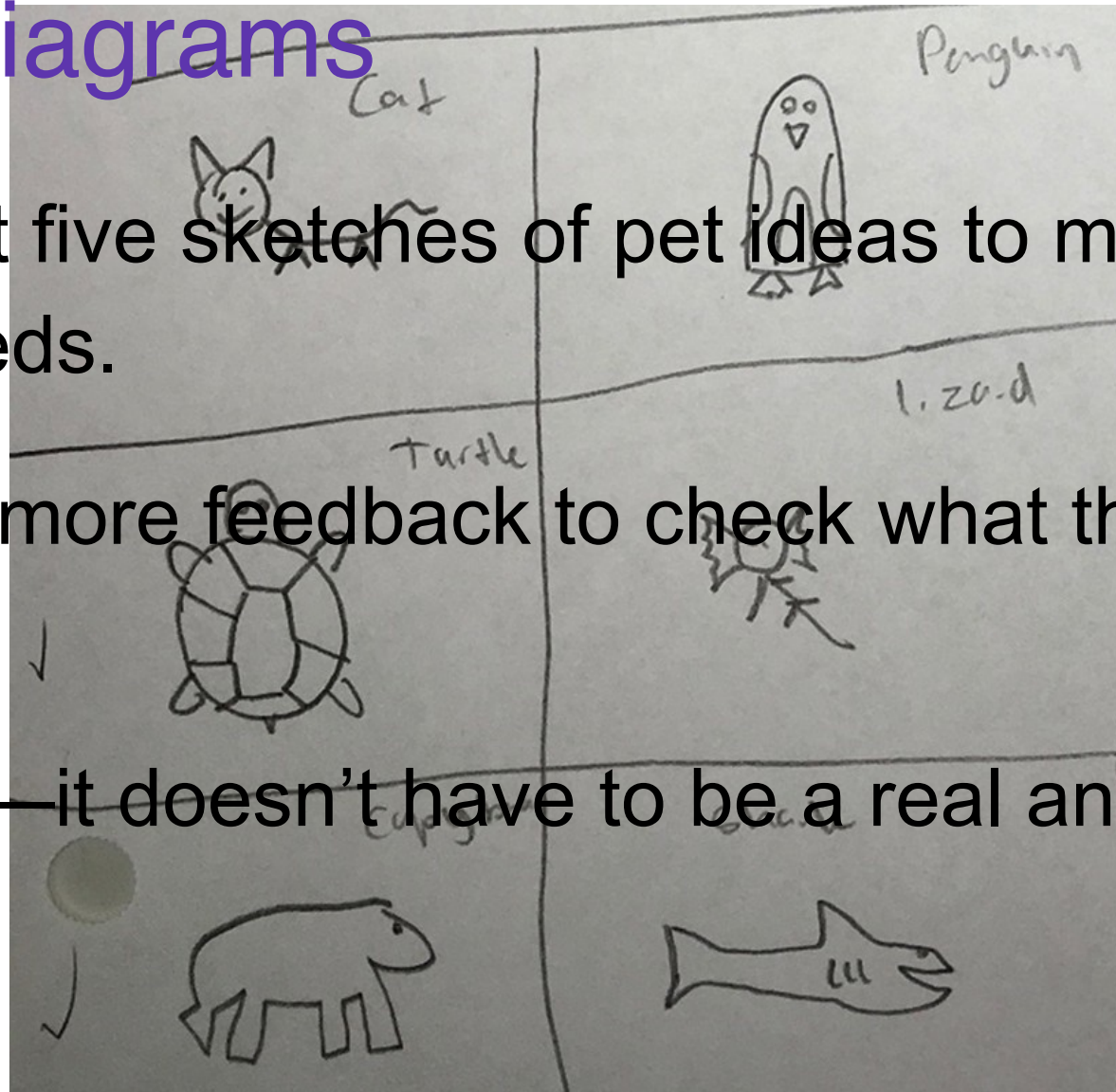
Suggested questions:

- Do you have a pet? What is it?
- If you don't, what kind of pet would you like? Why?
- What do you like and dislike about your pet?
- Is there anything you wish your pet could do? Why?
- What kind of activities would you do with your pet?
- How would you care for it? Where would it live?
- Tell me about your ideal pet.



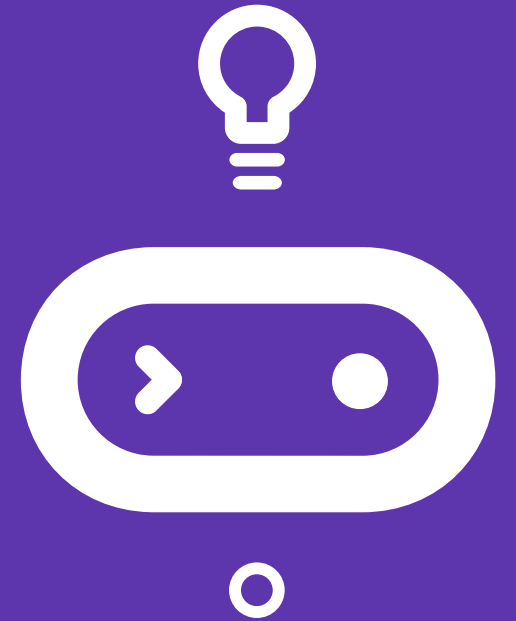
Prototype diagrams

- Draw at least five sketches of pet ideas to meet your partner's needs.
- Then gather more feedback to check what they like or don't like.
- Get creative—it doesn't have to be a real animal!



Let's review

- What is design thinking?
- What's the first step in coding by design?



What we learned today

- Design thinking/coding by design
- Interviewing to understand a need
- Definition statements and prototype diagrams



Next time: Lesson B

- Use Microsoft MakeCode and micro:bit
- Assess our learning with a quiz



Lesson B

Introduction to Microsoft MakeCode

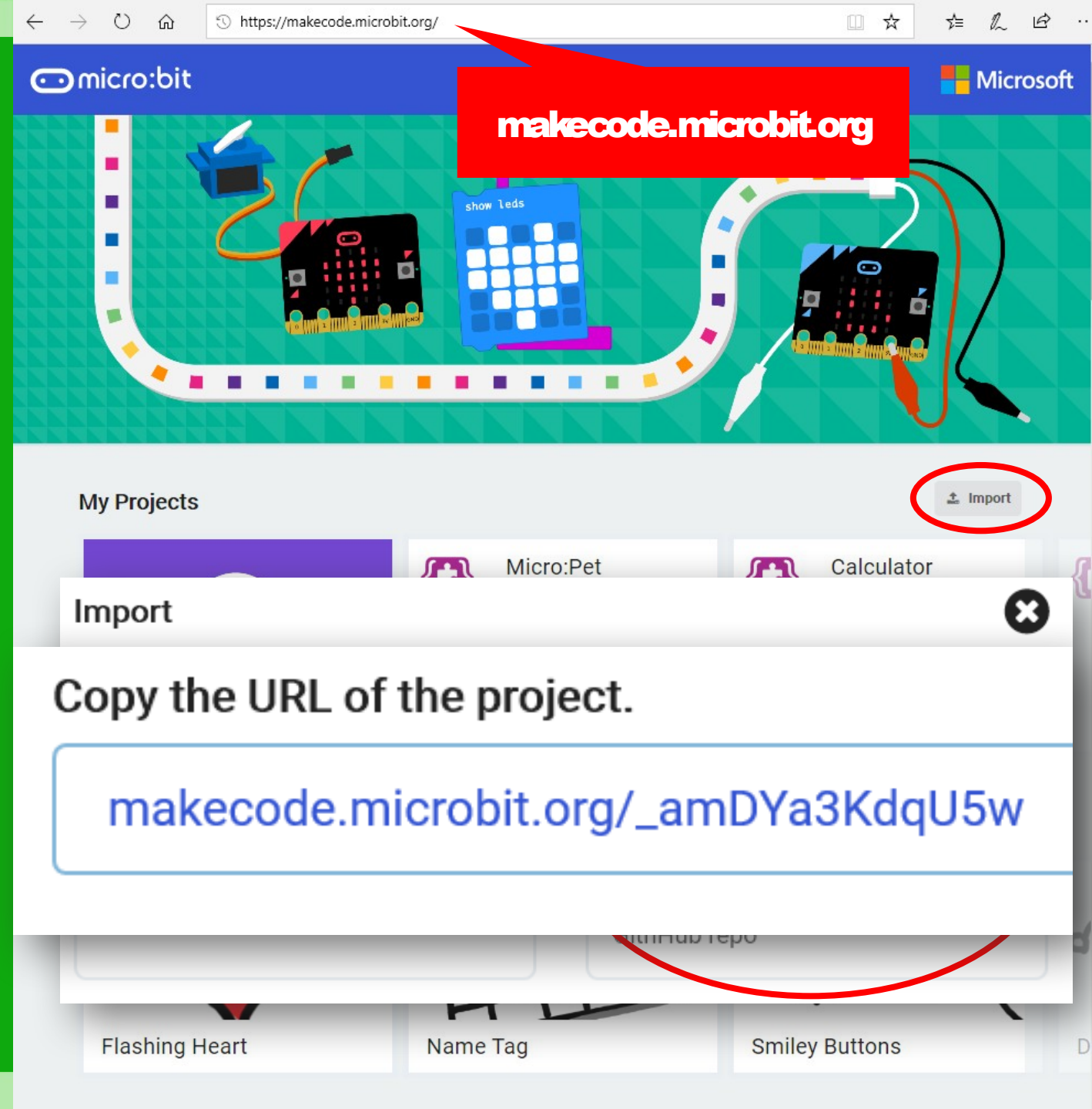


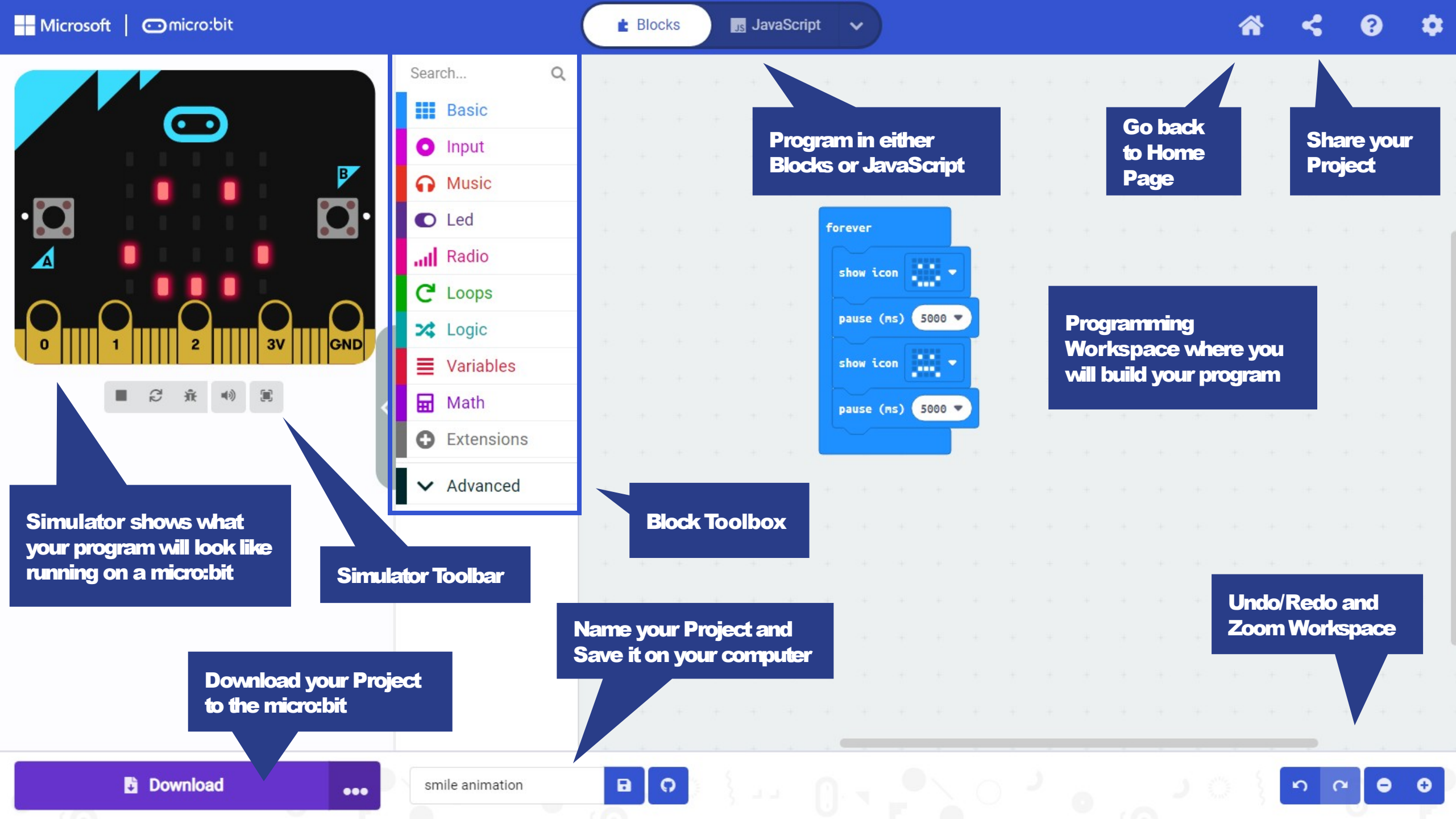
What we'll learn in Lesson B

- Open Microsoft MakeCode and upload a program
- Download a program to micro:bit
- How much we've already learned

Import a program in Microsoft MakeCode

1. In a browser, go to makecode.microbit.org or open the app.
2. Select “Import”.
3. Enter the URL:
makecode.microbit.org/_amDYa3KdqU5w
4. The smile animation project will open.





Blocks

JavaScript



Search...



Basic

Input

Music

Led

Radio

Loops

Logic

Variables

Math

Extensions

Advanced

Program in either
Blocks or JavaScript

Go back
to Home
Page

Share your
Project

Programming
Workspace where you
will build your program

Simulator shows what
your program will look like
running on a micro:bit

Simulator Toolbar

Block Toolbox

Name your Project and
Save it on your computer

Download your Project
to the micro:bit

Undo/Redo and
Zoom Workspace

Download

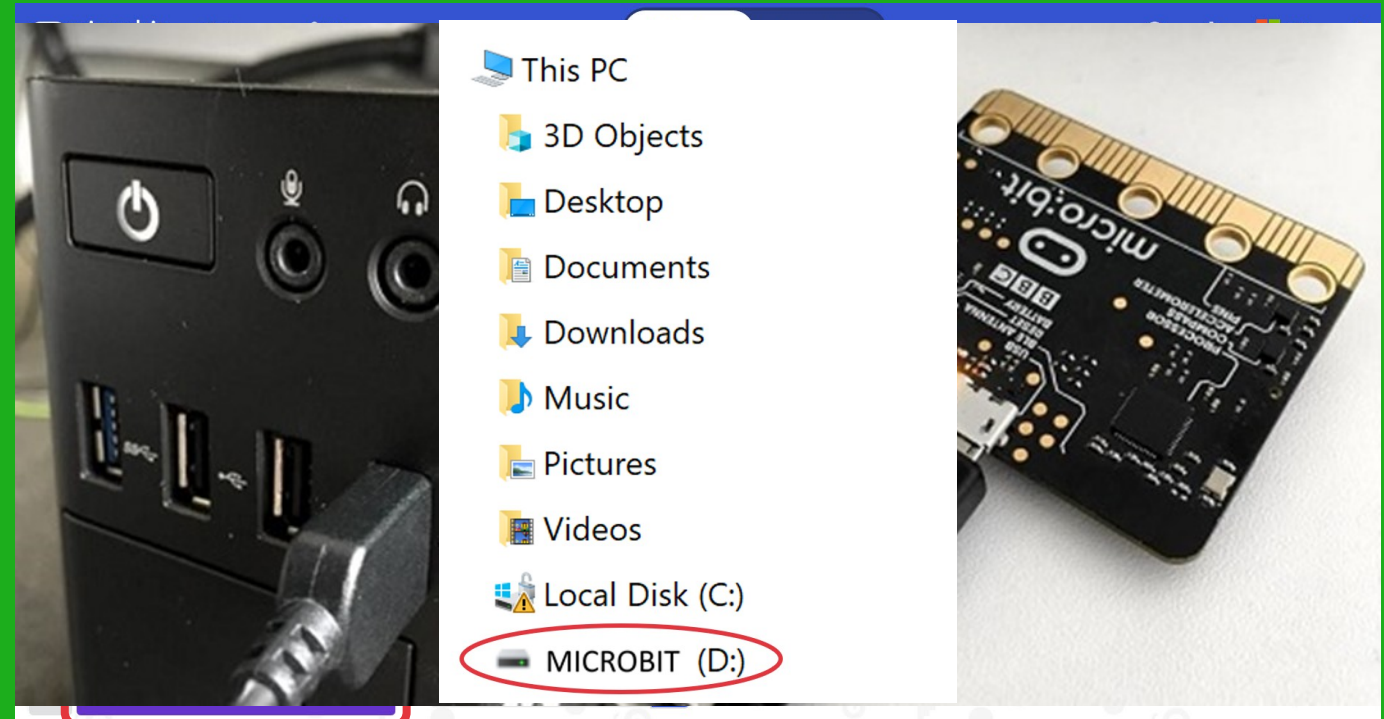


smile animation

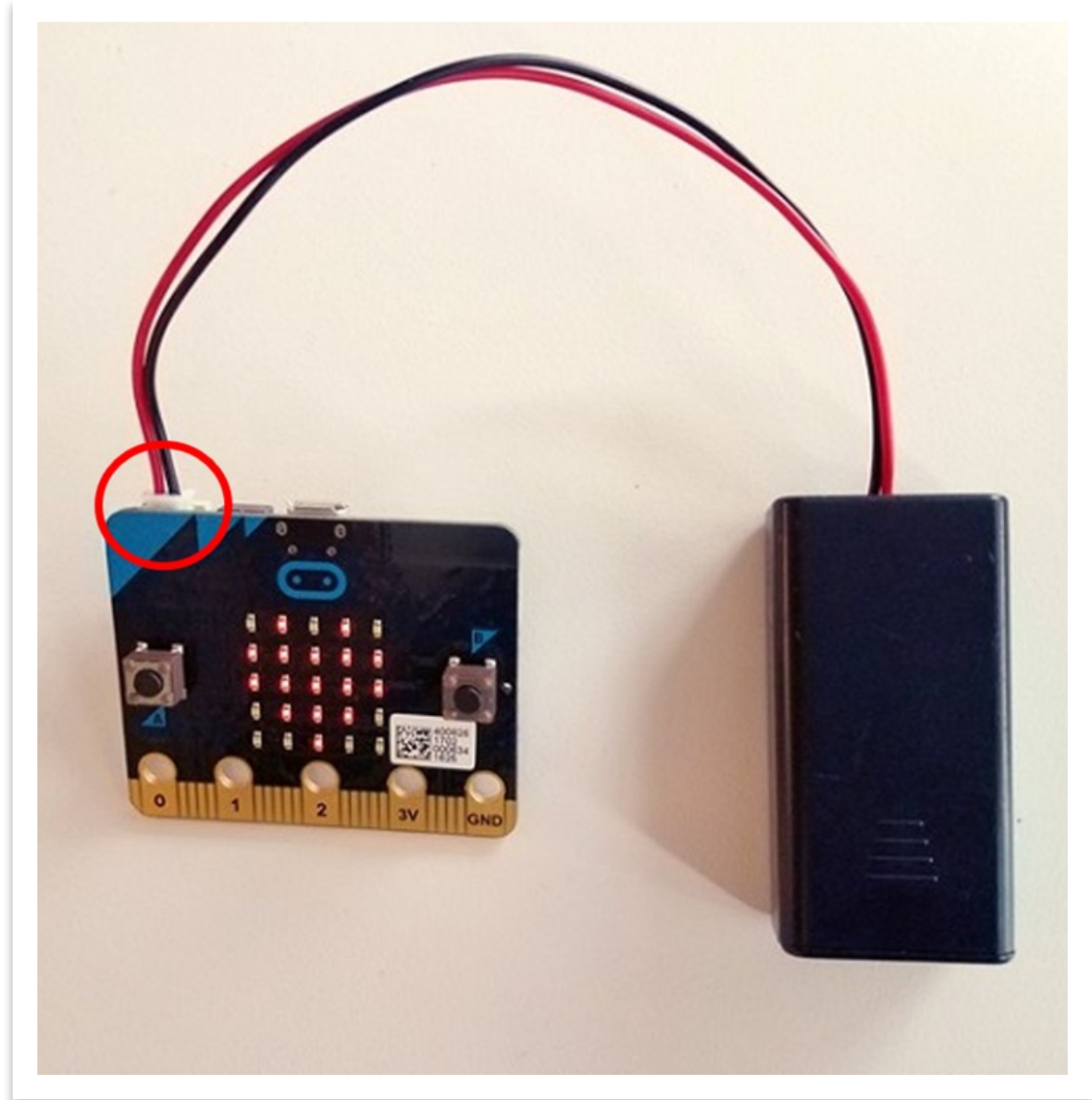


Download a MakeCode program to micro:bit

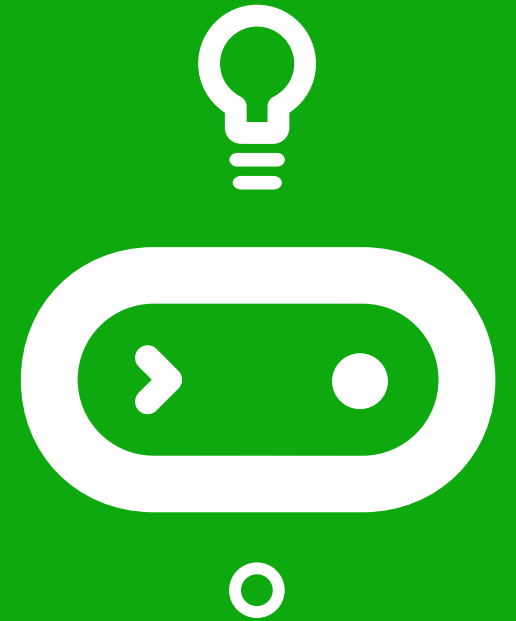
1. Connect the micro:bit to your computer using the micro-USB cable.
2. In the open MakeCode project, select **Download**.
3. Move the downloaded .hex file to the **MICROBIT drive**.



Connect the micro:bit to the battery pack



Time for a quiz!



What we learned in Lesson B

- Imported a program in MakeCode
- The layout in a MakeCode project
- Downloaded a program to the micro:bit
- Connected the micro:bit to the battery pack



Next time: Lesson C

- Design and build a micro:pet project



Lesson C

micro:pet project

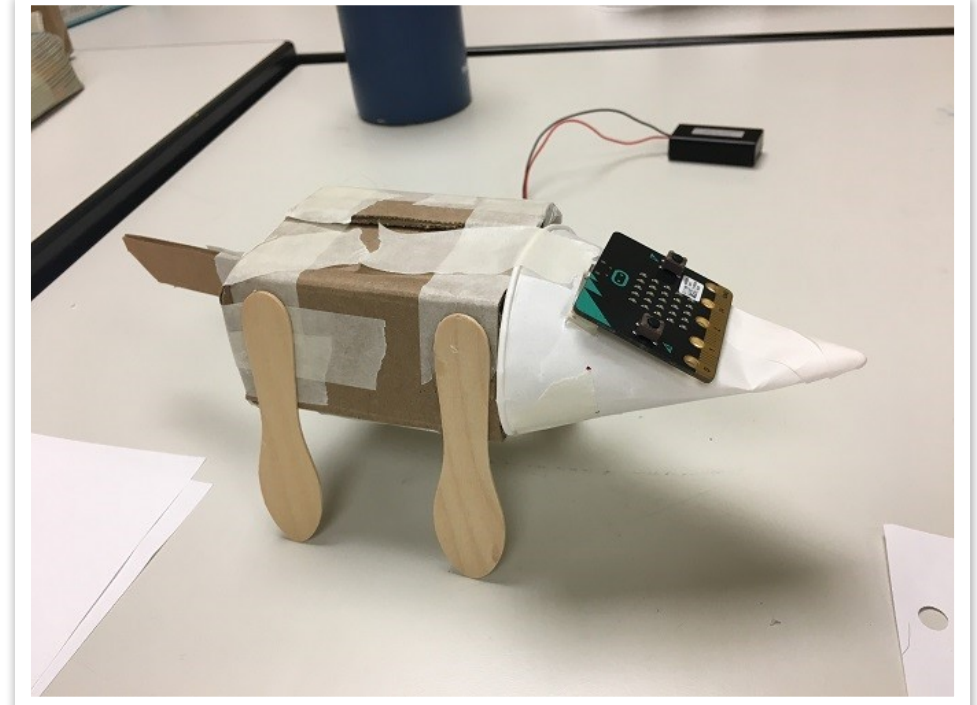


What we'll learn in Lesson C

- Design and build a micro:pet

Project expectations

- Review your interview notes, design statement, and prototypes, then think about:
 - Will it be an animal, plant, robot, bug?
Something else?
 - Will it have any moving parts?
 - If it moves, how can it hold the micro:bit securely?
- Build a micro:pet that:
 - Matches your partner's needs.
 - Supports the micro:bit (so you can easily turn it on and off) and battery pack.
- Don't glue or draw on the micro:bit!

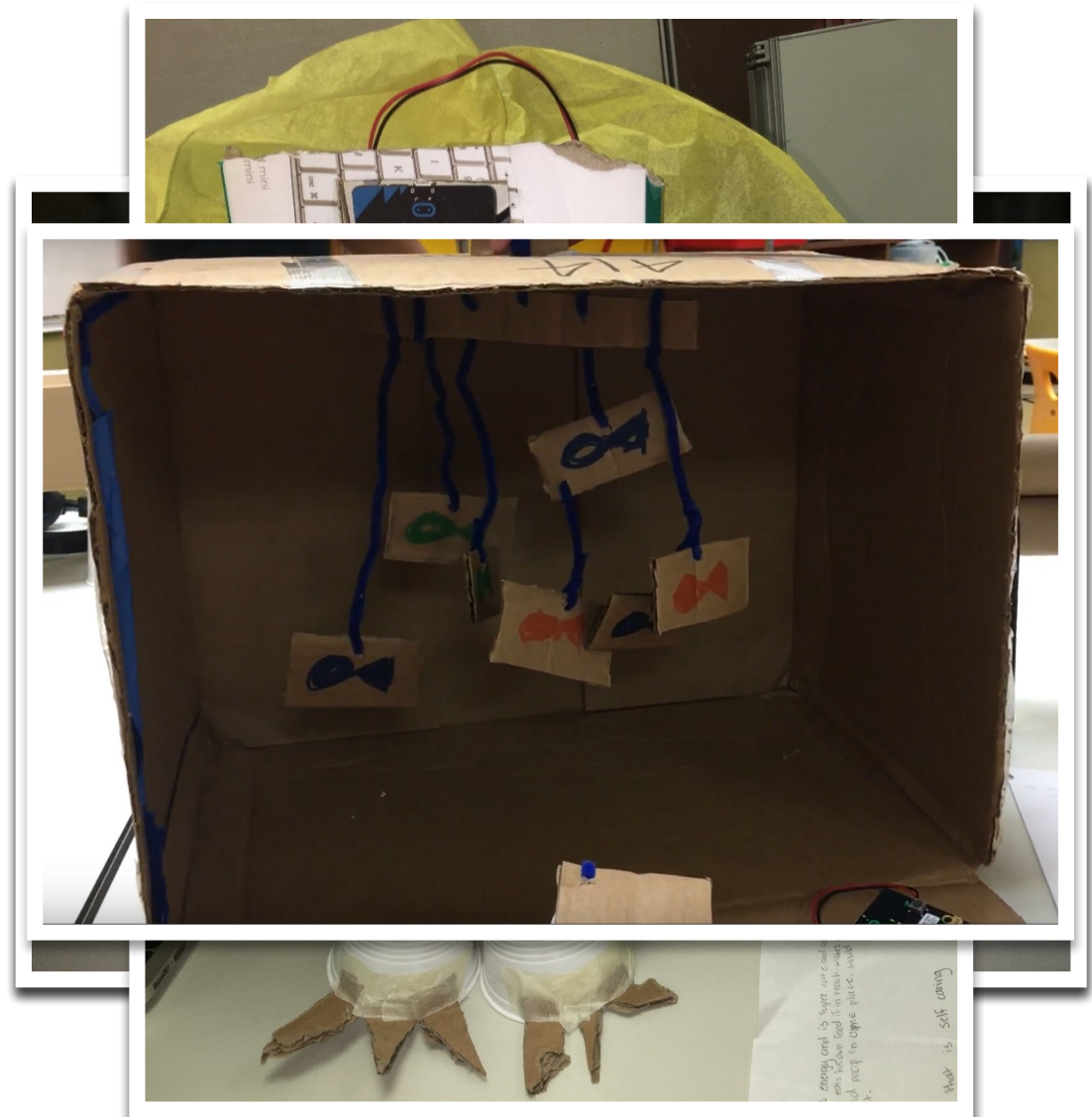


Other project ideas

Get creative and use your imagination:

- A pink piggy
- A ladybug
- A caterpillar
- A fox
- A robot
- Even a fish tank

Select to see
it in action!



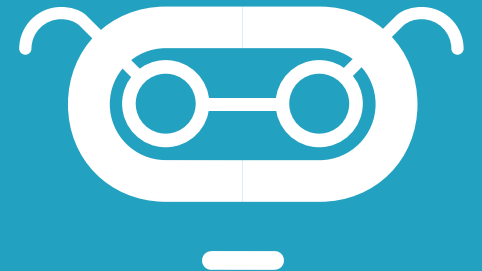
Modification (mod) ideas

- Find a way to make part of the animal move.
- Give your animal a natural habitat.
- Create a way to carry your animal.
- Create an animal that reacts when you pet or move it.



Reflection Diary

- Summarize the feedback you got from your partner.
- How would you revise your design?
- What was it like to have someone designing a pet for you?
- Was it a pet you would have enjoyed? Why or why not?
- What advice did you give them that might help them redesign?
- What was it like to interview your partner?
- What was it like to be listened to?
- What surprised you about the process of designing the micro:pet?
- Describe a difficult point in the process of designing the micro:pet and explain how you resolved it.



What we learned in Lesson C

- Turned information we learned in interviews into a design for a micro:pet
- Created the micro:pet

